

Shared UI State and Host Enforcement: An Empirical Technical Note on Three Agent-UI Configurations

Eduardo Arana

Arananet

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Abstract

Reused identifiers produce different retained states under the shared and separated mappings of three concrete tested configurations: json-render, A2UI, and MCP Apps with their local adapters and hosts. In a fixed-order write probe, Node assertions establish replacement of the json-render summary and retention of both summaries in separate MCP instances. The MCP browser fixture executes only the two summary writes, omitting Node’s preceding detail writes; it does not validate the same full S2 scenario. Executed browser assertions show finance’s headline visible and risk’s absent in the A2UI surface. The tested MCP host rejects a model-only tool call when its enforcing handler is installed; the permissive variant forwards that call. Interpretation must therefore name the adapter mapping, namespace/topology, and enforcement site together with the SDK or protocol. This empirical technical note with private local evidence audits an existing deterministic TypeScript harness across surface handoff, fixed-order identifier collision, and action round-trip. A claim-to-check mapping distinguishes supported, partial, and proposed claims; routing to every intended control owner remains partial, and a matched-topology intervention remains proposed, not performed. The evidence supports configuration-specific findings, not isolated protocol causality or rankings. No LLM experiment or performance measurement is reported. Human scientific and editorial review remains pending.

Browser provenance gap: the retained browser evidence predates the S2 rename. It cannot substantiate the post-rename source; fresh built browser execution is required. All executed DOM observations below refer to the pre-rename fixture.

1 Introduction

The existing harness exposes two configuration-dependent findings. Reusing a summary identifier replaces a value in json-render’s shared namespace, whereas the MCP adapter’s separate instances retain both values. A2UI’s shared mapping records the later writer; executed browser assertions confirm finance’s headline is visible and risk’s is absent in that surface. Separately, the tested MCP host rejects a model-only call with an enforcing handler but forwards it in the permissive variant. A comparison must identify the adapter mapping, namespace/topology, and enforcement site alongside the SDK or protocol: those choices are part of the observed system, not incidental implementation details.

This empirical technical note with private local evidence examines three concrete tested configurations, not three protocols in isolation. Its contribution is an auditable account of a small deterministic harness: research questions, a falsifiable hypothesis, bounded findings, an evidence map, and a local reproduction procedure. UI descriptions, execution environments, and authorization decisions operate at different layers; collapsing those layers would misattribute namespace choices or host policies to a format. The retained comparison report is an artifact to audit, not independent scientific evidence. Its S2 identity and labels now name the fixed-order collision. Scope and validity limits below delimit the findings; no exhaustive novelty claim is made.

2 Research questions

RQ1: Where do the existing adapters place the output of a scripted surface handoff?

RQ2: What survives a fixed-order identifier-collision probe in the configured namespaces? Only the one specified sequential write order is observed; it cannot generalize to alternative orders, scheduling or concurrency.

RQ3: Which local bookkeeping and host decisions determine action routing and model-only tool rejection?

3 Methodology

This retrospective case study examines deterministic scripted agents, with no LLM experiment. The unit of comparison is an SDK/adaptor/host configuration. Even uniform scenario inputs could not isolate topology, adapter, or protocol causality. Browser MCP inputs are not identical to Node inputs. Three scenarios exercise three configurations, with an existing permissive-host variant for MCP Apps in S3.

3.1 A falsifiable local hypothesis

H1: In the four-write Node S2 fixture, json-render retains finance’s headline and MCP Apps retains both headlines in separate instances. A2UI’s browser fixture displays finance’s headline and no risk headline within its surface; its Node check establishes only finance writer bookkeeping. These are separate fixture-specific predictions, not a common full-scenario browser validation. H1 observes only the one specified sequential write order and cannot generalize to alternative orders, scheduling or concurrency. A snapshot or DOM assertion showing the risk headline instead of finance in a shared summary, or loss of either isolated headline, would contradict this hypothesis for the tested configuration.

The S2 script awaits each emission: risk detail, finance detail, risk summary, then finance summary. This is a fixed-order identifier-collision probe, not actual concurrency or a scheduling benchmark. The source, runner, test titles and scenario ID now name the fixed-order collision. The focused check uses the existing snapshot assertions and browser assertions, rather than treating the adapter’s `LOST` or `ENFORCED` labels as measurements. Fresh execution status is recorded separately from this source-based hypothesis.

3.2 Configurations and versions

The manifest specifies ranges. The lockfile and locally installed package metadata agree on the exact versions below. Package versions are not protocol schema versions. These are observations of this checkout, not recommendations or statements about the newest available releases.

Configuration	Manifest range	Lockfile and installed version
json-render core and React renderer	<code>^0.21.0</code> each	<code>0.21.0</code> each
A2UI web core and Lit renderer	<code>^0.11.0</code> each	<code>0.11.0</code> each
MCP Apps extension SDK	<code>^2.0.0</code>	<code>2.0.0</code>
MCP client, core, and server	<code>^2.0.0</code> each	<code>2.0.0</code> each
React and React DOM	<code>^19.2.3</code> each	<code>19.3.0</code> each
Lit	<code>^3.3.3</code>	<code>3.3.3</code>
Zod	<code>^4.2.0</code>	<code>4.6.5</code>

The A2UI adapter imports `@a2ui/web_core/v0_9` and emits `version: "v0.9"`. Its package version `0.11.0` does not mean that the experiment uses a `v0.11` protocol. MCP Apps SDK `2.0.0` likewise does not establish a wire protocol version. The extension specification and SDK are separate references; no exact upstream specification commit or negotiated wire-version value was retained for this paper. Live repository documentation was inspected on 21 September 2026 and may differ from release-time documentation. The local lockfile fixes the tested package resolution, not the mutable web references.

The json-render adapter maintains one `Spec` per conceptual surface and maps logical block, row, and control IDs to element keys. It calls `applySpecPatch` for element additions/replacements and also directly updates root children. Therefore, its patch log does not capture every mutation. Its catalog validates emitted specs; `writtenBy` is an ordinary catalog prop and `lastWriter` is adapter bookkeeping. Neither is authenticated provenance.

The A2UI adapter validates messages using `A2uiMessageSchema` and drives `MessageProcessor`. Its component IDs and data paths derive from block IDs. A shared `rootChildren` collection coordinates the surface root. Node uses a schema-only harness catalog; the browser uses the Lit basic catalog. Snapshot titles and writer names partly come from adapter maps, while row values come from the data model. Surface addressing is not authorization: an address identifies a target but does not establish a caller's permission.

The MCP Apps adapter creates one server per agent and one app instance per surface/agent pair. Its Node path uses real SDK clients, servers, apps, and bridges connected by in-memory transports. Its snapshot is reconstructed from adapter-held blocks, not from a browser DOM. The browser host mounts an iframe with `sandbox="allow-scripts"` for each agent and uses `PostMessageTransport`. This topology is an implementation choice. Connection identity is not universally agent identity: the mapping works here because the host establishes one server per named agent. A server could aggregate multiple agents in a different application.

Figure 1 locates the shared state, local ownership mapping, and enforcing handler in these configurations. The dashed path denotes the existing permissive variant, not a second enforcement mechanism.

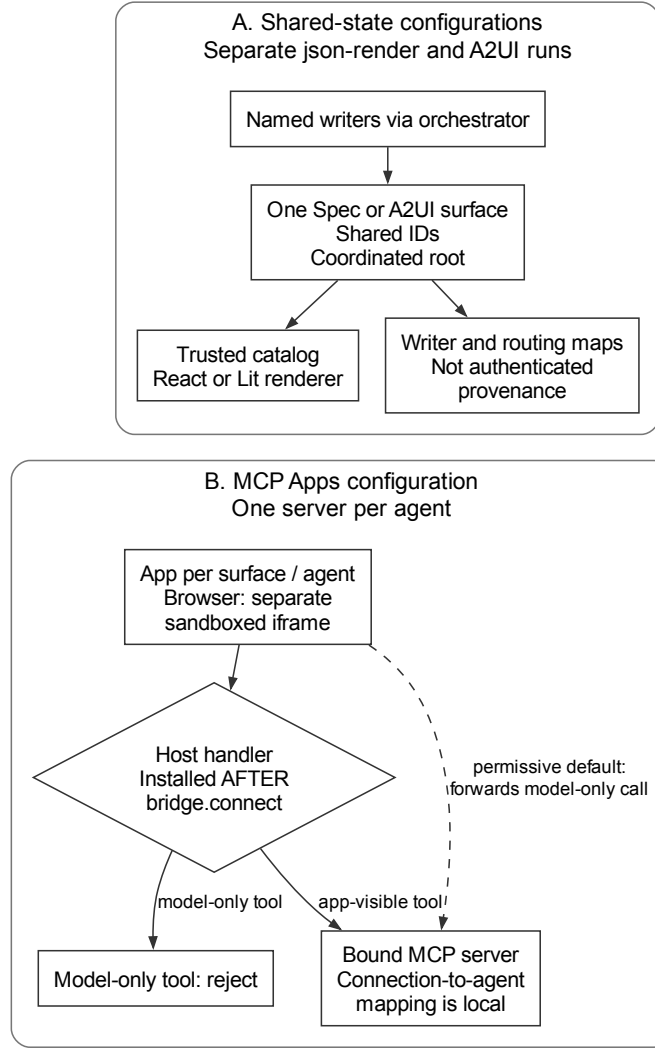


Figure 1: Configured state and enforcement boundaries. Panel A summarizes two separate declarative-renderer runs; panel B shows the app-to-server call path for each MCP instance. Writer maps are bookkeeping. In the tested enforcing variant, the installed host handler rejects the model-only call; the permissive variant forwards it. Original source-derived illustration, not a fresh browser observation.

3.3 Scenario design and evidence collection

S1 opens a trip surface, emits a planner itinerary, hands off to booking, and emits a reservation. S2 performs the fixed-order writes specified in H1. S3 emits basket and payment controls and activates ordinary and privileged actions. The amounts and booking operations are fixtures: the MCP server records calls and returns text; it does not charge money. The three scenarios are fixed examples, not a sample drawn from a population of agent workflows.

The runner constructs fresh adapters per scenario and configuration. The orchestrator is a scripted stand-in with named agents and a control-to-agent routing table, not an autonomous model-driven system. We inspect the existing tests and rerun them where local prerequisites permit; no scenarios or adapters are added or modified.

The report generator takes adapter-authored classifications and uses first-recording-wins selection for each capability and configuration. It does not independently measure the meaning of `SUPPORTED`, `LOST`, `ENFORCED`, or `NOT_EXPRESSIBLE`. Repeated labels do not constitute independent observations. Timestamp normalization makes its Markdown stable but does not validate its interpretation. A freshness check establishes correspondence between report and generator, not the truth of every sentence in the report.

We therefore distinguish classifications from assertions on snapshots, messages, tool responses/refusals, and rendered elements. The conformance tests validate the S1 A2UI messages and json-render catalog output, examine recorded patches, and check report and image-link freshness. They do not establish full specification conformance. The A2UI deletion message is not exercised in that S1 check. The json-render patch assertions do not cover its direct root mutation.

Specifically, `Tracer.record` stores the outcome supplied by the adapter; `outcomeOf` returns the first matching entry. In scenario tests, json-render’s S3 confirmation check and MCP’s permissive S3 check assert these labels only. MCP’s S2 `ENFORCED/NOT_EXPRESSIBLE` labels do not measure collision prevention or impossibility of composition; separate assertions check the two retained headline values. A2UI S2 checks labels and the writer map, not Node-rendered text. The bridge tests instead exercise calls and assert actual returned content or rejection plus refusal records. None of these labels is an empirical score.

Browser evidence has an additional qualification: the MCP browser host contains its own fixture script. In S2 it renders the two summaries without the two uncontested detail blocks used by the Node scenario: Node awaits risk-detail, finance-detail, risk/summary, finance/summary; browser MCP awaits risk/summary, finance/summary only. Its DOM assertions establish those browser-specific summary values, not validation of the same full Node S2 scenario. Retained screenshots alone cannot establish that a fresh browser run passed.

4 Bounded observed results

4.1 RQ1: Handoff and placement

Existing S1 assertions check a single `trip` snapshot region containing both `itinerary` and `reservation` in json-render and A2UI. MCP Apps checks separate `trip::planner` and `trip::booking` regions. The browser assertions check both blocks within one React tree or A2UI surface, and separate MCP iframes. These checks support a placement observation under the chosen mappings. They do not prove that MCP Apps cannot represent an aggregated UI or that either declarative format provides authenticated cross-agent ownership.

4.2 RQ2: Fixed-order identifier collision

These results observe only the one specified sequential write order and cannot generalize to alternative orders, scheduling or concurrency. Browser results are historical pre-rename observations; post-rename browser verification is pending.

The focused local S2 rerun passed its three selected tests; ten non-S2 tests in that file were skipped by the filter. The json-render assertion checks the finance writer and headline, plus survival

of the two uncontested blocks. The A2UI Node assertion checks the finance writer from adapter bookkeeping; its browser assertion checks headline text. MCP snapshot assertions check both headlines in different instances. H1 has fixture-specific support from these Node checks and the historical pre-rename browser run on 22 September 2026 UTC (21 September locally). A2UI's browser assertion checks finance's headline visible and risk's headline absent within the surface. It does not assert an exact summary node count, despite its test title. The raw browser report records 18 passed tests: nine render assertions and nine screenshot tests. The MCP browser fixture omits Node's risk-detail and finance-detail writes. Its two summary assertions are a separate browser result, not corroboration of the full four-write Node S2 or support across a common shared fixture.

The replacement is consistent with reusing a slot in these shared namespaces. It is not an observed scheduling race. The inspected adapter records a changed writer; this trace is not an audit of all conflict signals available in a protocol or SDK. Namespacing, ownership checks, arbitration, or shared aggregation could change the outcome; none is evaluated as an intervention here.

Figure 2 separates the four awaited writes from the retained summary states. The arrows express script order, not elapsed time or competing schedules.

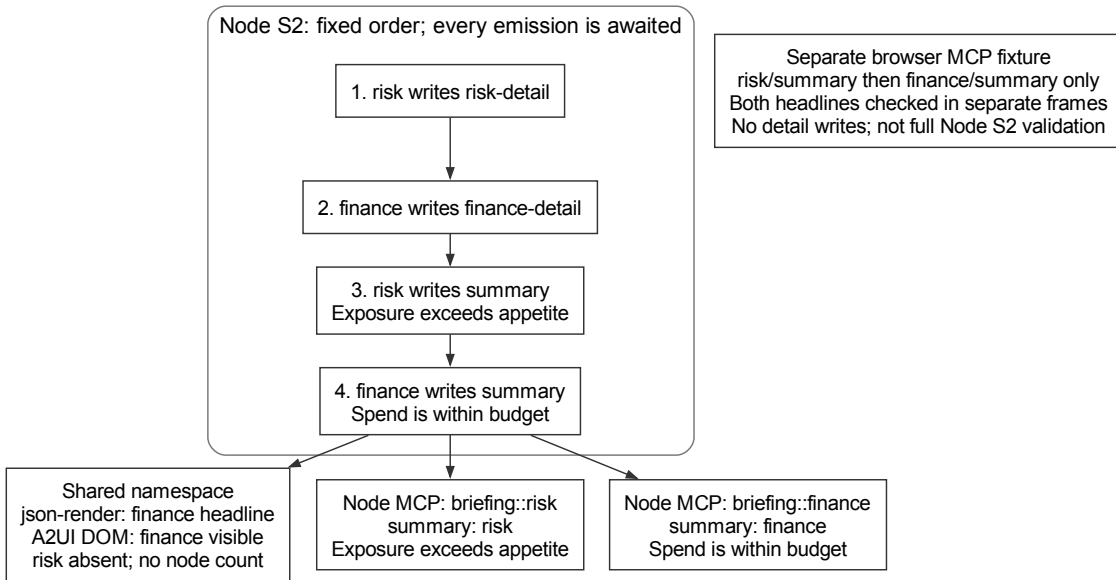


Figure 2: S2 fixed-order collision: the four-write sequence is the Node fixture. Node assertions check the json-render headline and both MCP headlines. Executed A2UI DOM assertions check finance's headline visible and risk's absent (C3, supported), not an exact node count. The separate MCP browser fixture executes only risk/summary then finance/summary, omitting both detail writes; it does not validate full Node S2. The two MCP summaries belong to separate instances, not a merged surface. Arrows indicate fixed script order, not time or scheduling.

4.3 RQ3: Routing and host decisions

In the shared-state configurations, ordinary action routing uses the orchestrator’s control table. Tests inspect routing entries and deliveries, and the A2UI action envelope carries surface/component fields and a timestamp without an agent field. These payload observations do not rule out an application-supplied identity envelope. Encoding a name in context or props would still require trusted binding before it could authorize anything.

json-render’s privileged control includes `confirm` metadata. The Node adapter constructs an event and classifies that metadata; it does not run an independent authorization service. The browser suite clicks the ordinary json-render action, not the privileged confirmation flow. Confirmation UI is not independent authorization, and the existing assertions do not establish complete consent behavior or protection against an omitted confirmation binding. A2UI has no privileged gate implemented in this adapter; that is not a proof that its hosts or trusted catalog components cannot impose one.

For MCP Apps, the host installs its enforcing `oncalltool` handler after `bridge.connect`. The handler looks up the tool and rejects model-only visibility. The bridge tests check successful app-visible calls, rejection messages and refusal records, and successful model-only calls when enforcement is disabled. The permissive host variant forwards that call. This is evidence about the tested SDK and handler installation order, not every MCP host or SDK release. The inspected extension specification assigns visibility enforcement to the host [R4].

Routing evidence is narrower than some trace labels suggest. For actions, the Node MCP adapter selects the first instance for a conceptual surface. Its helper is `findInstanceForSurface`. It does not find the agent owning the clicked control. The routing test only requires some delivery without the fallback table. Thus it cannot prove correct owner routing for every S3 control. Browser S3 instead clicks inside the cart and payments frames separately. We retain this distinction and leave the existing behavior unchanged.

5 Claim-evidence mapping

Here **supported** means a bounded statement has a matching inspected assertion or direct source observation, with run status separately recorded. **Partial** means a relevant assertion exists but leaves a material inference untested. **Proposed** means a future check has not been performed. These terms are this paper’s evidence statuses, distinct from the adapter outcome vocabulary.

Claim	Status	Evidence and falsifier
C1: S1 yields one shared region or two MCP instances in this topology.	Supported	S1 snapshot region and block assertions; contradicted by missing blocks or different region counts.
C2: Four-write Node S2 overwrites the json-render summary and retains both MCP summaries.	Supported	Node headline assertions; contradicted by different retained values. Browser MCP checks a separate two-write fixture, not full Node S2.

Claim	Status	Evidence and falsifier
C3: S2’s A2UI shared summary displays finance’s headline.	Supported	Pre-rename executed browser assertions check finance visible and risk absent within the surface; opposite visibility falsifies this outcome. Current-source browser verification is pending. No exact node count is asserted.
C4: S2 is sequential, with no scheduling test.	Supported	Four awaited emissions in the S2 source; contradicted by an overlapping scheduler in the executed path.
C5: The enforcing MCP handler rejects the model-only call, while the permissive variant forwards it.	Supported	Bridge rejection/refusal and successful-result assertions; contradicted by reversed or identical behavior.
C6: Every MCP S3 control reaches its intended agent.	Partial	Existing test checks some table-free delivery; first-instance selection prevents this stronger inference.
C7: json-render includes confirmation metadata, with consent enforcement incompletely checked.	Partial	Adapter source includes metadata; the scenario test asserts its classification only, not independent policy or privileged browser consent.
C8: Matrix labels are adapter classifications, not independent measurements.	Supported	Source shows first-recording-wins selection; freshness checks correspondence only.
C9: A matched-topology intervention isolates protocol causality.	Proposed	Pending controlled comparison; no new experiment is reported.

The linked scenario, bridge, conformance, and browser test files in Methodology and Results are the check locations for C1–C8. This mapping exposes uneven coverage rather than assigning a numeric score across unlike checks.

For C1–C3 and the tested browser actions, the renamed render spec supplies the assertions and the screenshot spec supplies nine captures. The JSON report, terminal output, commands, and source provenance identify the pre-rename tests only. Their hashes no longer match the current browser tests; the old raw logs cannot bind the post-rename fixture. A new built browser run is required before packaging current-source verification. C6 remains partial: successful browser actions in its distinct fixture do not repair the Node first-instance routing gap.

6 Related work

The json-render repository describes catalog-defined components and actions rendered from structured specifications [R1]. A2UI similarly separates declarative UI data from client implementation and describes a trusted component catalog [R2]. These are meaningful security properties: constraining requested components and handlers can limit what untrusted UI data asks the client to run. They do not by themselves establish per-writer authorization in our shared namespace. Conversely, our collision example is not evidence of blanket insecurity in declarative rendering.

The MCP Apps repository describes tool-associated UI resources and a host bridge [R3]. Its

specification describes visibility checks and sandbox/CSP obligations [R4]. Those obligations are not proof that this harness checks all of them. In particular, iframe sandbox attributes and successful bridge tests do not establish exhaustive CSP enforcement, transport security, or resistance to malicious content. Declarative catalogs and sandboxed app delivery are compatible layers: a hosted application could use a declarative renderer internally. No integration experiment is reported here.

The repository `besanson/sarc-authority-derivation` is acknowledged only as an unverified methodological pointer, not as inspected scientific evidence. Direct retrieval of its revision, README, REPRODUCTION.md, NOVELTY.md, and named v0.6.6 manuscript was unavailable in this environment. No revision, prose, results, or reproduction credit from that project is asserted here. Reference review remains pending; this discussion is selective, not an exhaustive literature or novelty review.

7 Threats to validity

Internal validity. Identical logical inputs leave adapter bookkeeping, shared-root coordination, naming conventions, host enforcement, catalogs, rendering frameworks, transports, and one-server-per-agent MCP topology as confounders. The default/permissive MCP comparison narrows one host-policy question, but it is not a factorial comparison across all configurations.

Construct validity. A region count is a property of the adapter snapshot; it is not a usability measure. A writer prop is not authenticated provenance. Addressability is not authority. A confirmation binding is not an authorization decision. Adapter labels and some scenario assertions share their interpretation, creating a risk of circular validation. The independent aspects of the checks are the inspected contents, response/refusal assertions, and DOM expectations; they are still authored within this repository, not by independent evaluators.

Execution and coverage. Node transports do not test browser isolation. The MCP browser fixtures differ from the Node scenario. Historical screenshots remain retained artifacts; the nine new captures are separately archived and do not establish historical reproduction. The suite does not comprehensively test adversarial inputs, identity forgery, transport authentication, CSP, cross-server policy, authorization revocation, or arbitrary concurrent schedules. No completed security review is claimed.

External and temporal validity. Three designed scenarios, fixed fixture data, and the pinned installed packages cannot establish population effects or general protocol superiority. Later SDKs, other renderers, different server topologies, and stricter application policies can behave differently. Mutable upstream documentation is contextual evidence, not a version-pinned release archive. No latency, throughput, cost, user study, or model capability was measured.

8 Reproducibility

The reproduction guide gives commands and expected semantic outcomes. The publicly clonable harness does not include the ignored paper inputs. The retained private review archive and its text manifest are stale after the S2 rename. They remain unchanged; regeneration is blocked until new browser execution matches current source and build hashes. Neither a public nor an independent paper build has been demonstrated. The local execution record separates successful commands, failures, unavailable prerequisites, and retained artifacts. This earlier editorial revision started from HEAD `b5d06eafcd4cf9cb0ed49131add55cbdecdbd4b7c`. Current packaging records its own HEAD and file hashes without resetting external changes. The earlier execution record identifies its

own historical base. The paper package, build scripts, and paper tests remain locally ignored and uncommitted. Consequently, Git status and OpenSpec’s tracked/nonignored input fingerprint do not identify the complete paper workspace. Separate paper checks and SHA-256 manifests record the local manuscript, tooling, assets, and generated outputs.

Semantic reproducibility means obtaining the specified regions, values, events, and refusal behavior under the same configuration. Byte identity is a separate question. A2UI action timestamps vary; report generation normalizes those timestamps. PDF metadata and toolchain resources can also vary. We check generated LaTeX against fresh Pandoc output under the recorded toolchain, not PDF byte identity across machines. The Markdown manuscript is the only editable paper source; LaTeX and PDF are generated derivatives.

This session is an agent-assisted local rerun of existing checks. It establishes neither independent-person reproduction nor human approval. The previous session’s reported dependency installation is not a fresh `npm ci` execution in this record. Missing checks and failed attempts remain visible rather than being counted as passes. OpenSpec command success cannot prove every acceptance criterion.

The complete local Vitest run passed 24 tests across three files and TypeScript checking passed. The standalone `report:check` command failed when `tsx` attempted to create an IPC listener (`EPERM`); the report freshness assertion inside the passing Vitest suite is separate evidence, not a successful standalone command. Earlier browser attempts failed before tests: a sandbox listener returned `EPERM`, and the later managed preview timed out after 60000 ms. The retained outer-terminal run then reused an explicitly started IPv4 preview and passed both existing browser specs (18 tests in 8.9 s), without source or configuration changes. New captures were archived separately and historical PNGs restored; this is now pre-rename evidence and cannot substantiate the post-rename fixture. The required new outer-terminal browser run has not occurred. Pandoc produced LaTeX and its freshness check passed. The initial Tectonic invocation panicked internally (exit 101) in cached-only mode. A later local build used an explicit directory bundle assembled from already cached TeX resources and a smaller template, and produced the illustrated PDF without network access or shell escape. Original Mermaid diagrams were converted locally to vector figures through a restricted Mermaid-to-Graphviz renderer after Chromium-based rendering failed. These illustrations explain source and assertions; they add no experiments. The execution record retains the historical failures and current inspection evidence. Local compilation does not establish arXiv TeX Live compatibility or publication acceptance. Human review remains pending.

9 Conclusion

S2 observes only the one specified sequential write order and cannot generalize to alternative orders, scheduling or concurrency. The retained browser findings predate the rename; current browser-source provenance requires re-execution.

Across three concrete tested configurations, handoff placement and retained values reflect the chosen shared or separated mappings. The S2 assertions check `json-render`’s replaced summary and MCP’s two retained summaries; executed A2UI DOM assertions show finance’s headline visible and risk’s absent. H1 has only fixture-specific support: Node S2 has four writes, while browser MCP executes only risk/summary then finance/summary and cannot validate the full Node fixture. Neither fixture tests concurrency, scheduling, interleaving, or randomization. MCP model-only tool rejection depends on the tested enforcing host handler, while correct routing for every control remains only partially checked. Interpretation must name the adapter mapping, namespace/topology, and enforcement site together with the SDK or protocol. These results do not isolate protocol causality

or establish unavoidable incompatibility between composition and security. Integrations remain possible, and trusted catalogs retain meaningful security properties.

Future experiments are explicitly pending: matched namespaces and topology, controlled scheduling with overlapping writes, trusted identity envelopes, privileged confirmation interaction tests, and complete host-policy tests. They are suggestions for separate approved work, not results of this paper.

10 AI-assistance disclosure

An OpenAI Codex coding assistant inspected local source and accessible references, assisted with drafting and editorial revision, prepared source-derived figures and supporting documentation, added focused manuscript checks, and executed the recorded local validation commands. Human scientific and editorial review remains pending, including verification of claims, citations, limitations, and final presentation. The execution ledger is an agent-assisted record, not an author attestation or publication approval. AI-assisted preparation is separate from the deterministic harness: no LLM participates in its scenario runs.

11 References

R1. Vercel Labs. `json-render` repository and README. Inspected 21 September 2026; live main, upstream commit not retained. Local SDK context: `@json-render/core` and `@json-render/react` 0.21.0, including the installed core README and manifest. Used for catalog/renderer design, not a security proof.

R2. A2UI project. A2UI repository and README (the inspected `google/A2UI` URL redirects here). Inspected 21 September 2026; live main, upstream commit not retained. Local context: web core and Lit 0.11.0, installed web-core README, and adapter-selected v0.9 schema.

R3. Model Context Protocol contributors. MCP Apps SDK repository and README. Inspected 21 September 2026; live main, upstream commit not retained. Local context: extension, client, core, and server packages 2.0.0; extension README and manifests inspected. SDK package version is distinct from wire version.

R4. Model Context Protocol contributors. MCP Apps extension specification, `draft/apps.mdx`. Visibility and sandbox/CSP sections inspected 21 September 2026; mutable main path, no pinned specification commit. Used to distinguish host requirements from the assertions implemented here. The filename’s “draft” is retained as the source location, without claiming a release status from the SDK version.